

Thomas Evans

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EXPERIENCE

University of Alberta, Department of Family Medicine

Edmonton, AB

Software Developer

April 2025 – Present

- Building an analytics platform from zero infrastructure: SQL Server database, C# ASP .NET Core MVC application, Razor/JS frontend, Python ETL pipelines, and AI-assisted querying, serving multiple clinic sites.
- Designed and built a SQL Server system with temporal tables for tracking historical data changes and point-in-time analysis; built Python and SQL ETL/ELT pipelines to migrate legacy spreadsheet and SharePoint data.
- Designed and built custom report, dashboard, and chat user interfaces using Razor views, vanilla JavaScript, and HTML/CSS
- Designed an English-to-SQL conversion layer: used Gemini's API to generate SQL from executive natural-language input against a fully isolated read-only database replica; configured context-caching to reuse precomputed input tokens and avoid redundant schema retokenization.
- Secured the conversion layer with defense-in-depth: used ScriptDom to AST-parse AI-generated SQL and reject non-SELECT query patterns; hardened prompts to soft-secure against SQL antipatterns; applied least-privilege with permission-scoped connection string; stored schema metadata in Google Cloud Storage.
- Wrote technical documentation including full system specification, security policy, and formal requirements specifications with acceptance criteria and testing strategies.
- Pitched a platform to department executive leadership and secured approval to move from prototyping to production; led requirements gathering for 16 clinical end-users; conducted embedded site observations of clinical staff workflows.
- Developed ad-hoc internal tools using JavaScript and Python, ranging from data entry automation scripts with GUIs to a Flask and SQLite web application for clinical faculty management.

Research Assistant

May 2025 – Jul 2025

- Managed data collection, analysis, and study design for usability study comparing faculty reporting systems.
- Implemented PDF report generation and statistical analysis (ordinal logistic regression, sign tests, Cronbach's α , PCA).
- Adapted System Usability Scale for domain context; wrote technical report; lead author on poster (ACFP 2026); oral presenter at 2026 FM Research Day.

Program Assistant

Sep 2022 – April 2025

- Prototyped Python tool using Gemini API for data deidentification; automated administrative tasks with Python and JavaScript.

PROJECTS

Synesthesia

JavaScript, React. Front page Hacker News, Jan 2026. [Try it here](#)

Color-picker sonification tool. Answers the all-important, widely-asked question: 'What does a color-picker sound like?'

- Designed an algorithm mapping hexcodes to chords in the 24 tone-equal-temperament scale.
- Wrote bit-twiddling transformation logic to implement hexcode-to-chord morphisms.
- Implemented realtime audio synthesis with Attack-Decay-Sustain-Release envelopes using Tone.js.
- Used sound React patterns e.g. memos and refs to reuse Tone.js nodes, saving overhead on per-render regenerations.

TECHNICAL SKILLS

Languages: Python, C#, JavaScript/TypeScript, SQL, Rust, C, C++, R

Frameworks: .NET Core, Entity Framework, React, Flask

Infrastructure: SQL Server, PostgreSQL, SpacetimeDB, Redis, Kafka, Docker, Git, Linux

EDUCATION

University of Alberta

BSc, Specialization in Computing Science

Edmonton, AB

Part-time; Expected Fall 2027

Selected Coursework: Algorithms, Operating Systems, Parallel & Distributed Systems, Cryptography & Digital Privacy